

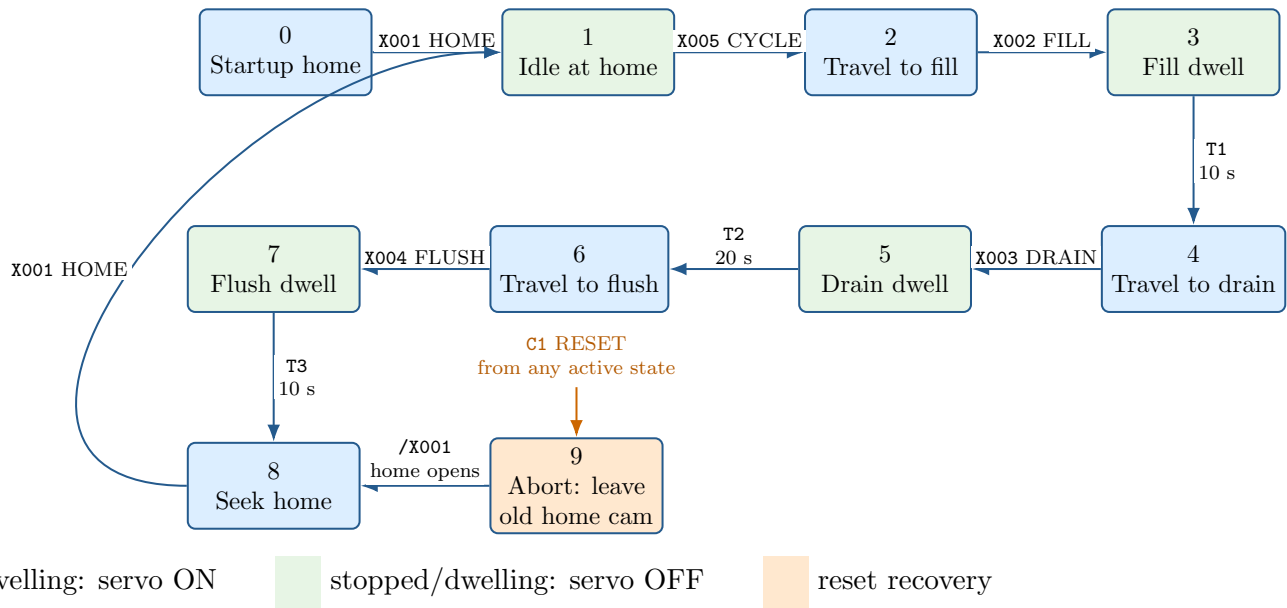
Project 4: Rotary Valve State Control

AutomationDirect CLICK PLC state diagram, timers, and ladder implementation

CLICK address assignment

Function	CLICK address	Type	Description
HOME	X001	Physical input	Home position switch
FILL	X002	Physical input	Fill position switch
DRAIN	X003	Physical input	Drain position switch
FLUSH	X004	Physical input	Flush position switch
CYCLE	X005	Physical input	Cycle-start pushbutton
SERVO	Y001	Physical output	Servo drive run/enable command
RESET	C1	Internal bit	Reset/abort request
STATE	DS1	Integer register	Current state, 0 through 9
POSITION	DS2	Integer register	0 home, 1 fill, 2 drain, 3 flush, 4 travelling

State diagram



State and output table

State	Meaning	Y001	Exit condition / next state
0	Startup homing	ON	HOME X001 → 1
1	Idle at home	OFF	CYCLE X005 → 2
2	Travel to fill	ON	FILL X002 → 3
3	Fill for 10 s	OFF	Timer bit T1 → 4
4	Travel to drain	ON	DRAIN X003 → 5
5	Drain for 20 s	OFF	Timer bit T2 → 6
6	Travel to flush	ON	FLUSH X004 → 7
7	Flush for 10 s	OFF	Timer bit T3 → 8
8	Return/search for home	ON	HOME X001 → 1
9	Abort; clear old HOME indication	ON	HOME opens (NC X001) → 8

CLICK ladder implementation

Initialization

CLICK DS registers are retentive by default. Initialize the state when the CPU enters RUN mode:

```
SC2 (First Scan)    --- COPY 0 to DS1
```

Use a Single Copy instruction with source constant 0 and destination DS1.

Normal transitions and dwell timers

Use Compare contacts for each expression such as $DS1 = 0$. Configure each timer as ON-delay with automatic reset when its enabling rung becomes false.

```
DS1 = 0   X001    --- COPY 1 to DS1
DS1 = 1   X005    --- COPY 2 to DS1
DS1 = 2   X002    --- COPY 3 to DS1

DS1 = 3    --- TMR T1, 10 seconds, ON-delay, automatic reset
DS1 = 3   T1     --- COPY 4 to DS1
DS1 = 4   X003    --- COPY 5 to DS1
DS1 = 5    --- TMR T2, 20 seconds, ON-delay, automatic reset
DS1 = 5   T2     --- COPY 6 to DS1
DS1 = 6   X004    --- COPY 7 to DS1
DS1 = 7    --- TMR T3, 10 seconds, ON-delay, automatic reset
DS1 = 7   T3     --- COPY 8 to DS1
```

Reset/abort recovery

First process the recovery transitions:

```
DS1 = 9   /X001    --- COPY 8 to DS1
DS1 = 8   X001     --- COPY 1 to DS1
```

Place the reset rung *after all state-transition rungs* so it has final control of DS1 during that scan:

```
C1 RESET    --- COPY 9 to DS1    (One Shot)
```

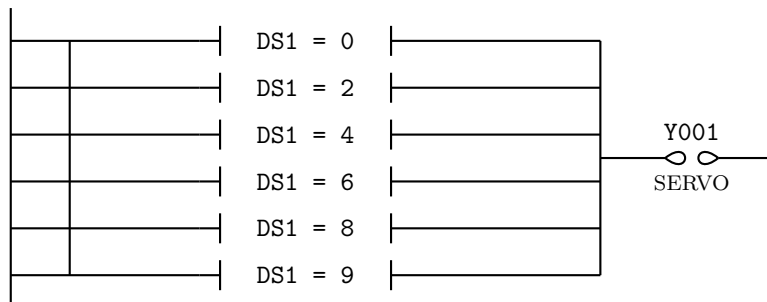
The One Shot option prevents a maintained reset signal from continuously forcing state 9. State 9 is needed because RESET may be initiated while the original HOME cam is still closed. The valve must leave that old HOME indication before state 8 accepts a new HOME signal.

Servo output

Use six parallel Compare-contact branches driving one physical output coil Y001:

$$Y001 = (DS1 = 0) \vee (DS1 = 2) \vee (DS1 = 4) \vee (DS1 = 6) \vee (DS1 = 8) \vee (DS1 = 9)$$

The complete output-activation rung is:



If any one of the six comparisons is true, the branch network has continuity and the OUT Y001 coil is energized. States 1, 3, 5, and 7 are not included, so Y001 is OFF while the valve is idle or dwelling.

Do not use multiple separate OUT Y001 coils. The branch network must terminate at one Y001 coil.

Position indication and commissioning

Position indication in DS2

Execute these rungs in the order shown. The always-on default reports travelling; an active position input later in the scan overwrites it.

```
SC1 (Always On)    --- COPY 4 to DS2
X001 HOME          --- COPY 0 to DS2
X002 FILL          --- COPY 1 to DS2
X003 DRAIN         --- COPY 2 to DS2
X004 FLUSH         --- COPY 3 to DS2
```

Only one position switch should normally be closed at a time. Add a fault indication if simultaneous switches must be detected.

Program termination

The final unconditional rung in the CLICK Main Program is:

```
--- END
```

Commissioning and safety notes

- Confirm that X001--X005 correspond to installed physical inputs in **Setup > System Configuration**.
- Confirm the exact CLICK CPU output type before wiring Y001: relay, sinking transistor, or sourcing transistor.
- Connect Y001 only to a compatible servo-drive run/enable input or an appropriately rated interface relay. Never connect a motor directly to a PLC output.
- Test initially with the motion command disconnected or replaced by an indicator lamp.
- Implement the emergency stop and personnel-safety functions in suitable hardwired or safety-rated circuitry, independent of ordinary PLC logic.
- Add a maximum travel timer and fault state so a failed position switch cannot command indefinite motion.